



FIRST SEMESTER: 2026-2027
COURSE HANDOUT

Date: 1st August 2026

In addition to part I (the General Handout for all courses, appended to the Timetable), this section provides further specific details about the course.

Course No : SNS F214
Course Title : Fundamentals of Solid-State Physics
Instructor-in-Charge : Jayendra N. Bandyopadhyay

1. Course Description: This course provides a comprehensive introduction to the fundamental concepts of solid-state physics with a particular emphasis on semiconductor materials and their applications. It covers crystal structures and defects, the quantum and statistical principles governing electron behavior in solids, the origin of energy bands, and the electronic properties of metals, semiconductors, and insulators. The course further introduces carrier transport mechanisms, including drift, diffusion, and the Hall effect, before discussing the thermal, magnetic, and optical properties of solids. Throughout the course, the underlying physical principles are connected to modern semiconductor devices such as LEDs, solar cells, photodetectors, and sensors, providing students with the essential foundation required for advanced studies in materials science, condensed matter physics, and semiconductor technology.

2. Scope & Objective of the Course: This course introduces the fundamental principles of solid-state physics that underpin the behavior of crystalline materials and modern semiconductor devices. It encompasses crystal structures, basic quantum and statistical concepts, electronic band theory, semiconductor physics, and the thermal, magnetic, and optical properties of solids, while emphasizing their relevance to contemporary electronic and optoelectronic technologies. The primary objective is to develop a clear understanding of the relationship between the atomic structure of solids and their physical properties, explain the quantum mechanical origin of electronic behavior in materials, introduce the principles governing charge transport in semiconductors, and establish the scientific foundation necessary for the study, analysis, and application of semiconductor materials and devices in advanced engineering and physics courses.

3. Course Learning Outcomes:

- (i) Describe crystal structures, Bravais lattices, crystal systems, common crystal structures, and Miller indices, and explain their significance in determining the physical properties of crystalline solids.
- (ii) Apply the fundamental principles of quantum mechanics to explain wave-particle duality, quantized energy levels, and the origin of electronic states in solids.
- (iii) Differentiate between Maxwell–Boltzmann, Bose–Einstein, and Fermi–Dirac statistics, and explain the statistical behavior of electrons and other particles in solids.
- (iv) Explain electrical conduction in solids using the Drude model and band theory, and relate density of states and effective mass to the electronic properties of materials.
- (v) Analyze the electrical properties of intrinsic and extrinsic semiconductors and explain carrier transport through drift, diffusion, and the Hall effect.



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(vi) Explain the thermal and magnetic properties of solids in terms of lattice vibrations, phonons, heat capacity, and magnetic ordering.

(vii) Explain the optical properties of semiconductors and relate them to the operation of optoelectronic devices, including LEDs, solar cells, photodetectors, and semiconductor sensors.

4. Textbooks:

(i) Charles Kittel, Introduction to Solid State Physics, 8th Ed., Wiley (2012) [TB1].

(ii) Arthur Beiser, Concepts of Modern Physics, 8th Ed., McGraw-Hill (2026) [TB2]

(iii) Donald A. Neamen, Semiconductor Physics and Devices, 4th Ed. (2007) [TB3]

5. Reference Books:

(i) Neil W. Ashcroft and N. David Mermin, Solid State Physics, Harcourt College Pub. (1976).

(ii) Kenneth S. Krane, Modern Physics, 4th Ed., Wiley (2021).

6. Course Plan:

Module No.	Lec. No.	Reference	Topics
1. Crystal Structure and Binding	1	TB1. Chap. 1	- Introduction to solids, Basics of crystal structure, Basis, Unit cell - Bravais lattices: Primitive and conventional cells - Fundamental types of lattices: 2D, 3D - Simple crystals: NaCl, Hexagonal closed pack, ZnS
	2	TB1. Chap. 1	
	3	TB1. Chap. 1	
	4-5	TB1. Chap. 1	
	6-7	TB1. Chap. 3	Crystals of inert gases, Ionic crystals, Covalent crystals
2. Basic Quantum Concepts	8-10	TB2. Chap. 5	- Wave equation, Schrödinger equation, Superposition, Expectation values - Solution of Schrödinger equation: Particle in a box, Finite potential well, tunneling
	11-12	TB2. Chap. 5	
3. Statistical Physics for Solids	13	TB2. Chap. 9	- Basics of statistical physics: Microstates, macrostates - Maxwell-Boltzmann (classical) statistics: Distinguishable particles - Quantum statistics: Bose-Einstein, Fermi-Dirac, and their applications
	14	TB2. Chap. 9	
	15-18	TB2. Chap. 9	
4. Electrons in Solids and Band Theory	19-21	Instructor Notes	- Classical free electron model; Electrical conduction: Drift velocity, mobility, conductivity; Need for quantum description - Band theory: Valence and Conduction bands, Kronig-Penney model - Classification of solids: Metals, Semiconductors, and Insulators - Conceptual introduction of density of states, Idea of effective mass
	22-23	TB1. Chap. 7	
	24	TB1. Chap. 7, TB2. Chap. 10	
	25-27	TB1. Chap. 5, TB3. Chap. 3	
5. Semiconductor Physics	28-29	TB3. Chap. 4	- Intrinsic semiconductors: Electron-hole pairs, carrier density, temperature effect - Extrinsic semiconductors: Donor impurities (n-type) and Acceptor impurities (p-type) - Carrier transport: Drift, diffusion; Hall effect
	30-31	TB3. Chap. 4	
	32-34	TB3. Chap. 5	



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6. Thermal and Magnetic Properties	35-36 37-38	TB1. Chap. 4, Chap. 5 TB1. Chap. 11, Chap.12	- Thermal properties: Lattice vibrations - Magnetic properties: Diamagnetism, Paramagnetism, and Ferromagnetism
7. Optical Properties and Applications	39-40	TB3. Chap. 14	Optical properties and applications: Photoconductivity, Solar cells, Photodetectors, LEDs

8. Evaluation Scheme:

Components	Duration	Marks (%)	Date & Time	Nature of component (Closed Book/ Open Book)
Mid-Semester test	90 mins.	30%	08/10 AN1	Closed Book
Class participation (Best 4 out of 8)	-	10%	Unannounced	Closed Book
Tutorial Tests (Best 3 out of 4)	30 mins.	20%	To be announced	Closed Book
Comprehensive Examination	180 mins.	40%	12/12 FN	Closed + Open Book

- **Attendance Policy:** Each student is expected to attend all classes and participate in all evaluation components.
- **NC Criteria:**
 - **30% of the average of the top three performers or 40% of the median, whichever is lower.**

9. Chamber Consultation Hour: Will be announced in the class. Also, feel free to email me to seek a prior appointment.

10. Notices: On Google Classroom.

11. Make-up Policy: Stringent. Only in the most genuine cases, as judged by the IC. Make-up will not be given for the tutorial tests.

Instructor-in-charge
Course No. SNS F214